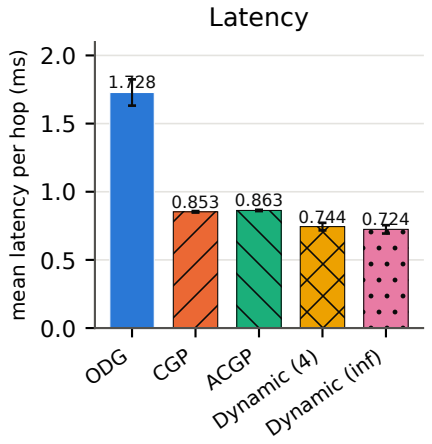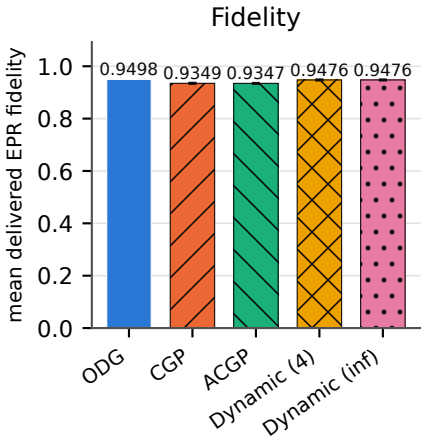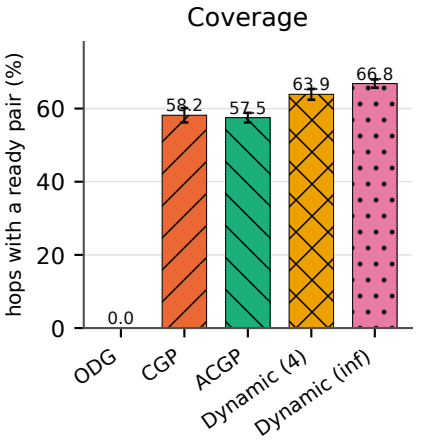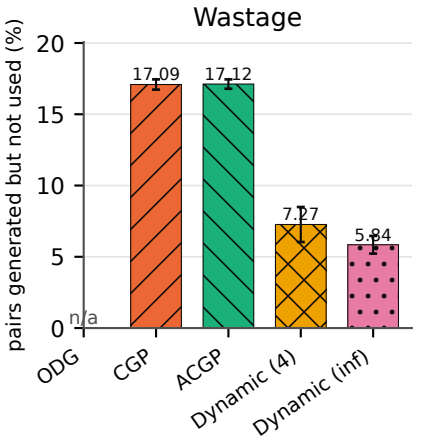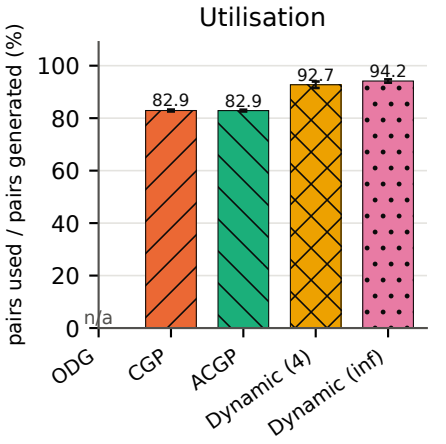BMW cardinality, history routing: 4 memories per core, all 4 for pre-generation